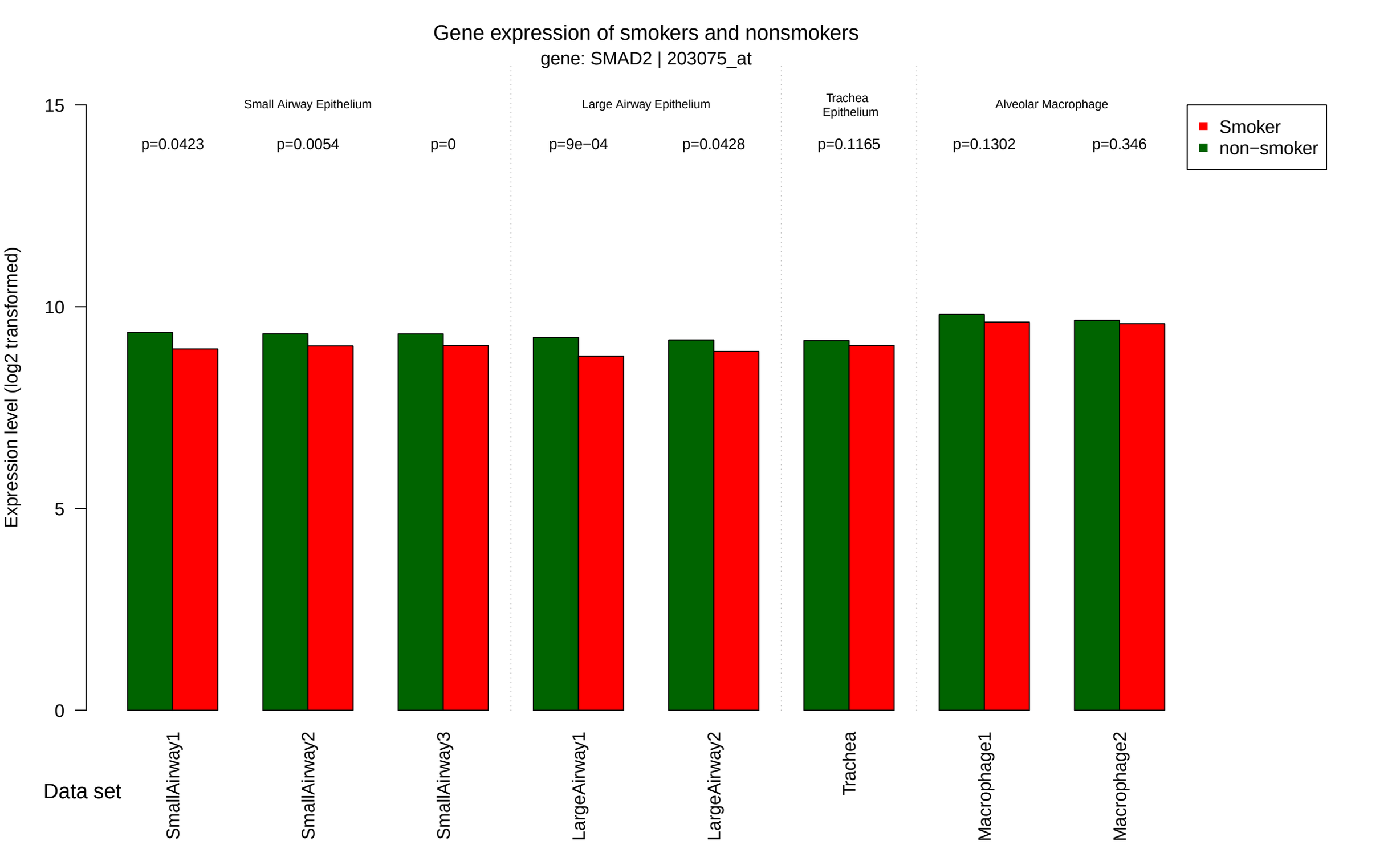
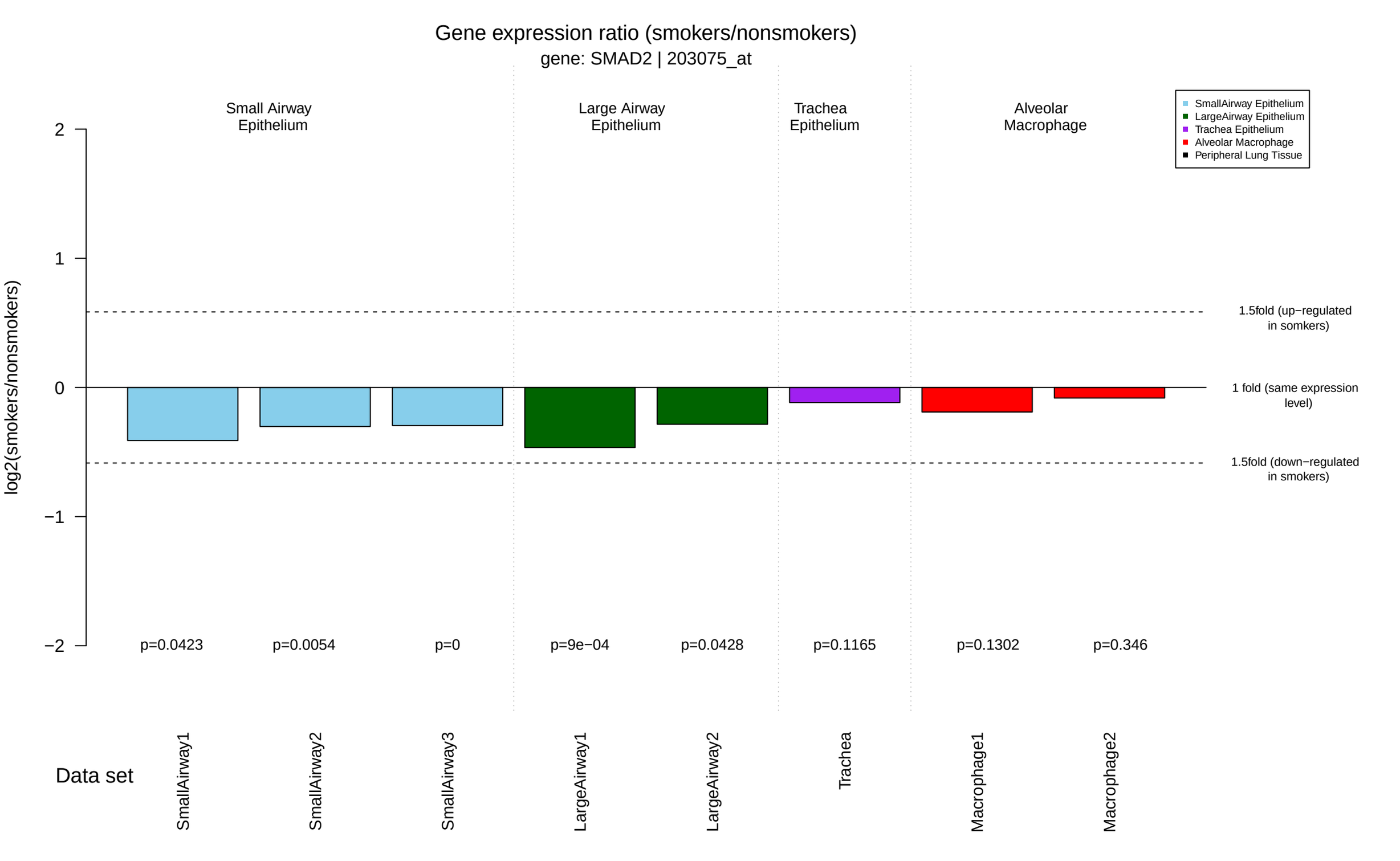
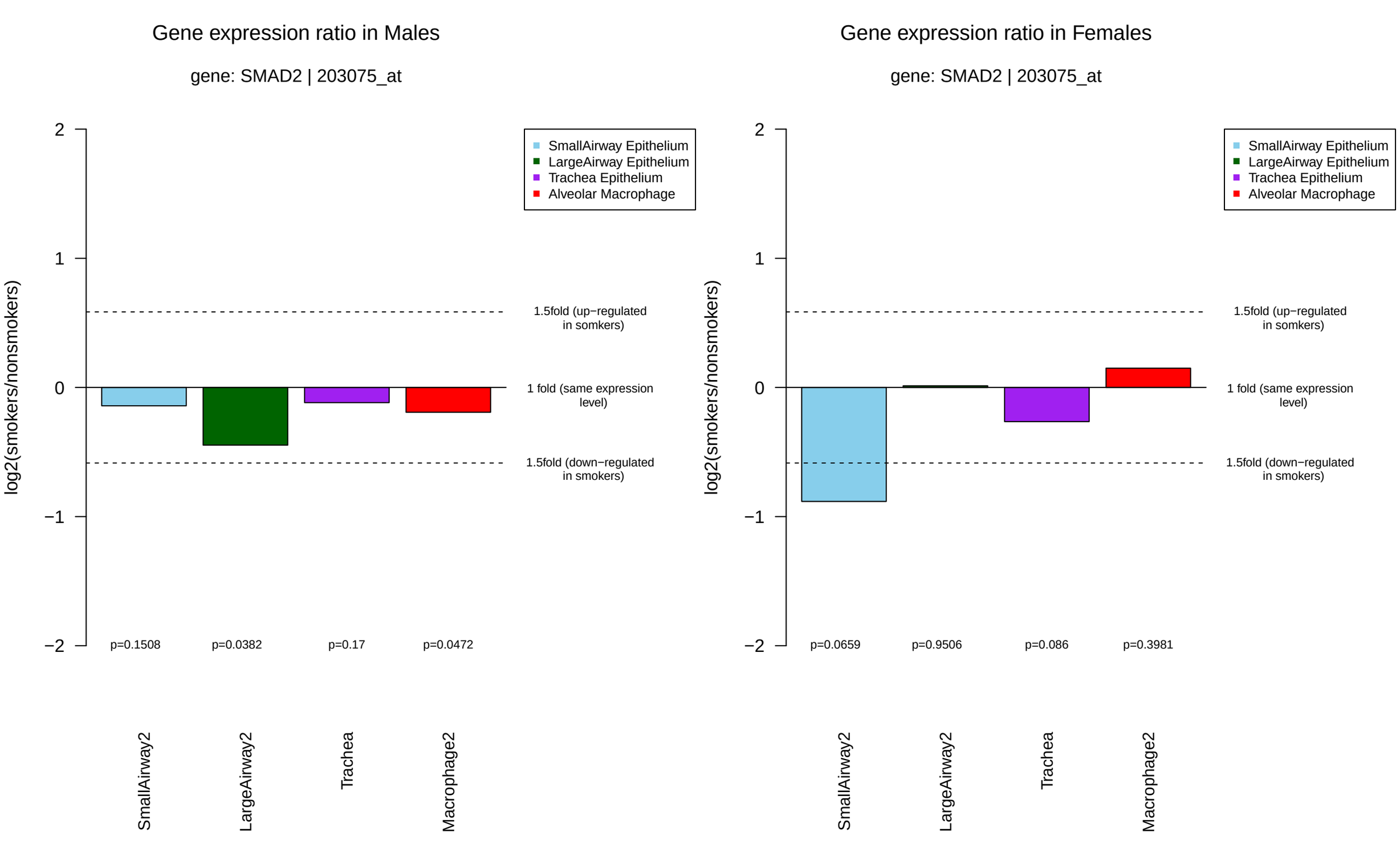
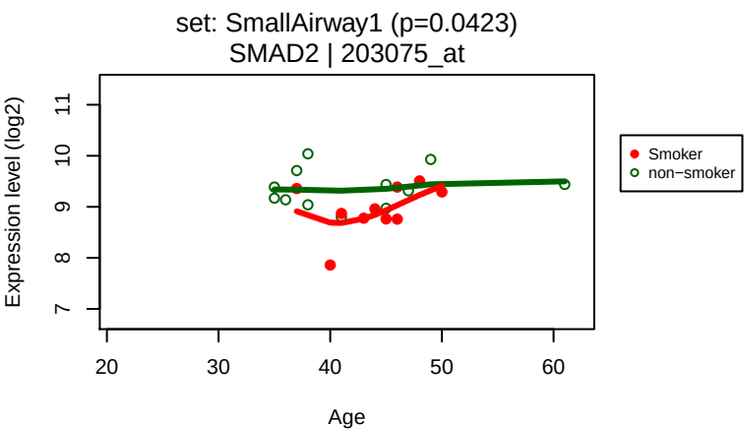
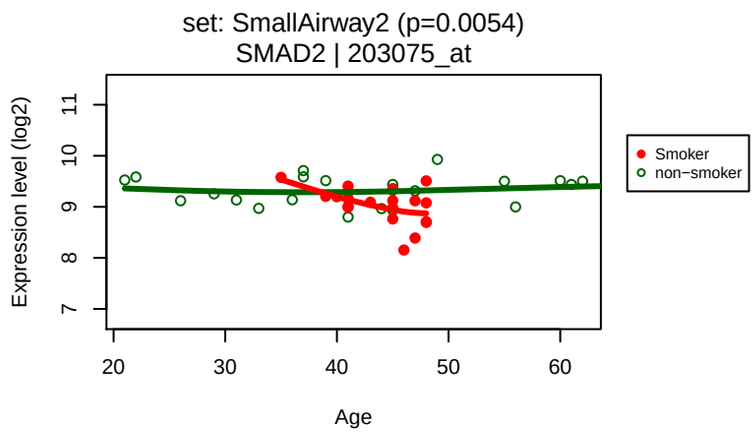
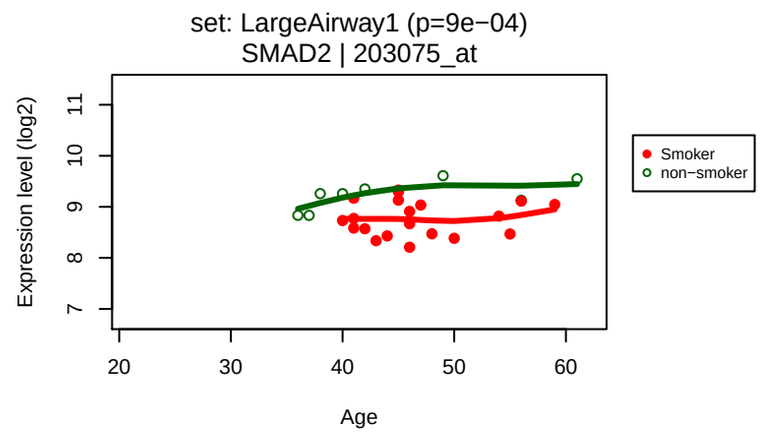

Gene expression of smokers and nonsmokers
gene: SMAD2 | 203075_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0423
p=0.0054
p=0
p=9e−04
p=0.0428
p=0.1165
p=0.1302
p=0.346
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

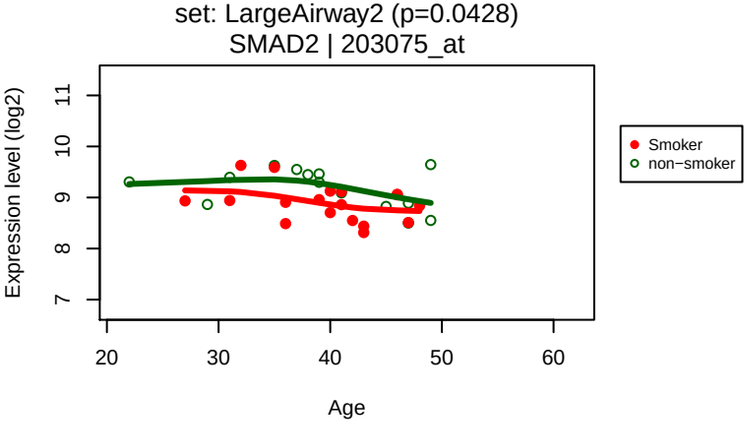

Gene expression ratio (smokers/nonsmokers)
gene: SMAD2 | 203075_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0423
p=0.0054
p=0
p=9e−04
p=0.0428
p=0.1165
p=0.1302
p=0.346
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

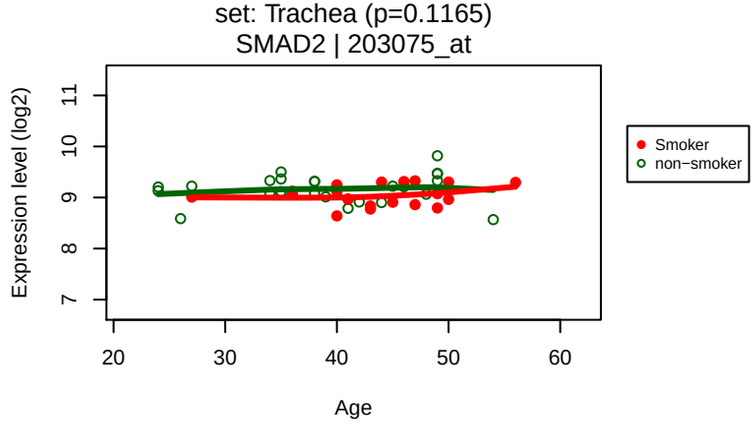

Gene expression ratio in Males
gene: SMAD2 | 203075_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1508
p=0.0382
p=0.17
p=0.0472
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD2 | 203075_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0659
p=0.9506
p=0.086
p=0.3981
SmallAirway2
LargeAirway2
Trachea
Macrophage2

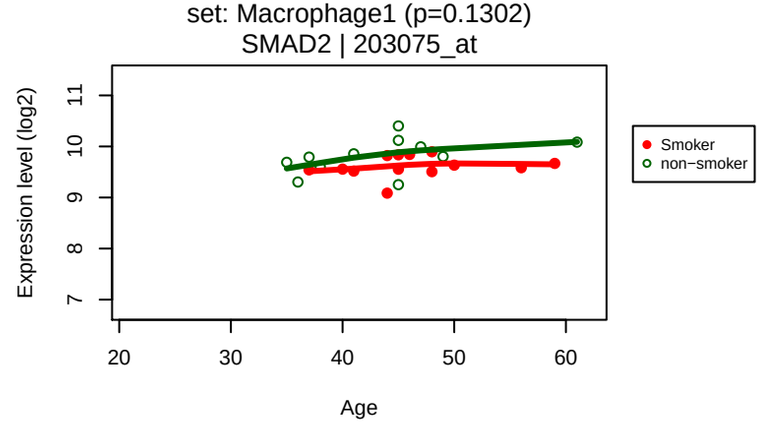

set: SmallAirway1 (p=0.0423)
SMAD2 | 203075_at
Expression level (log2)
Age
Smoker
non−smoker

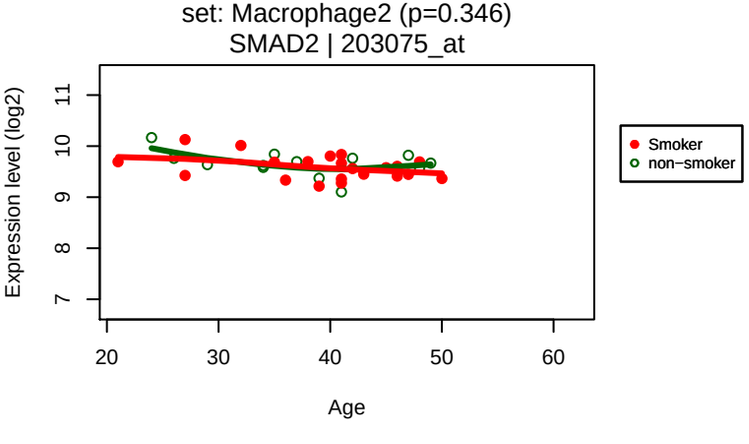

set: SmallAirway2 (p=0.0054)
SMAD2 | 203075_at
Expression level (log2)
Age
Smoker
non−smoker

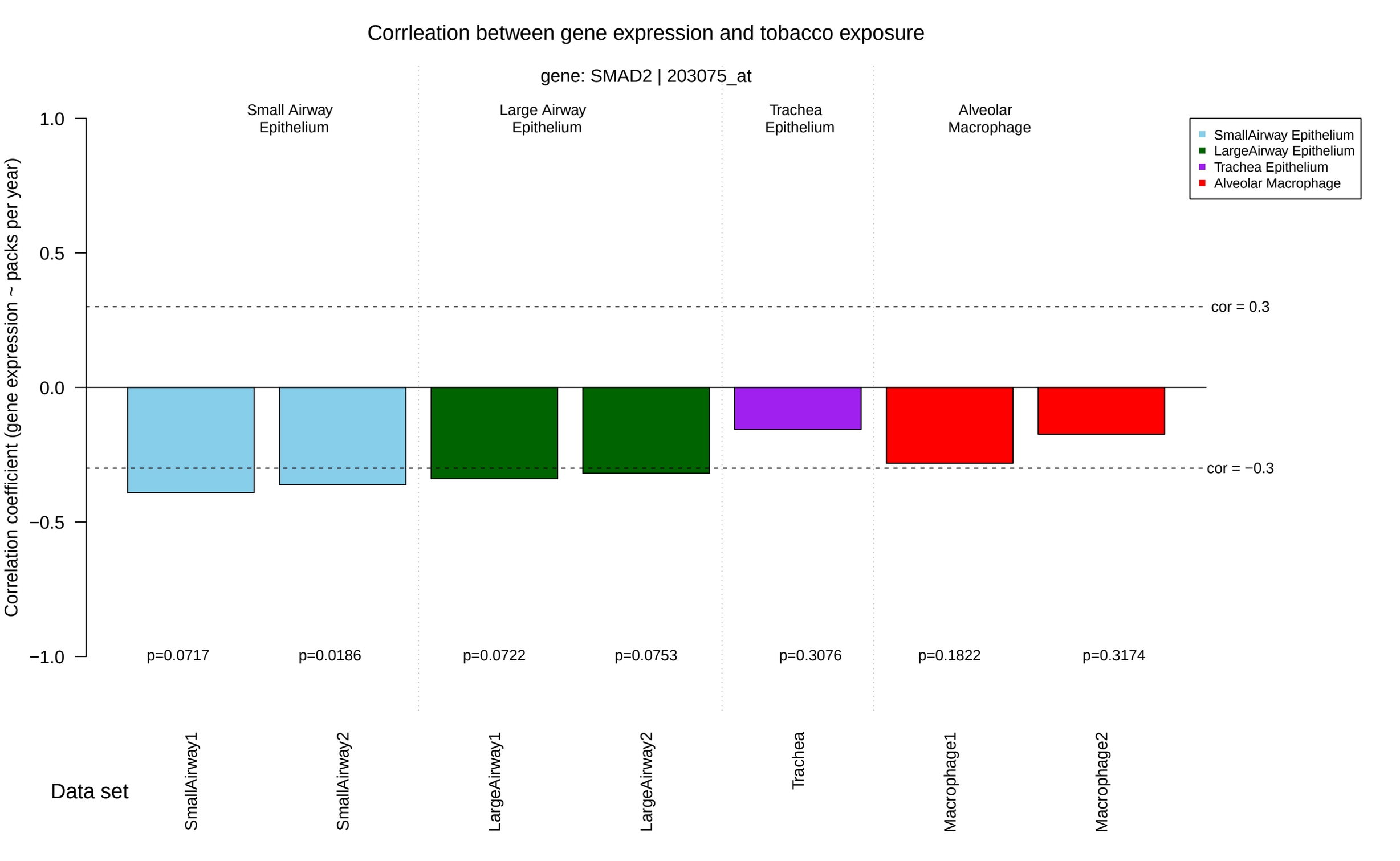

set: LargeAirway1 (p=9e−04)
SMAD2 | 203075_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0428)
SMAD2 | 203075_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1165)
SMAD2 | 203075_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1302)
SMAD2 | 203075_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.346)
SMAD2 | 203075_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD2 | 203075_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0717
p=0.0186
p=0.0722
p=0.0753
p=0.3076
p=0.1822
p=0.3174
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2